Mars Rover Sample Collection Planning

Analysis Report Using PDDL

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1. Introduction

An autonomous rover (Curiosity) on Mars must collect geological samples from six exploration sites and deliver them to the Base. The rover has two storage bays (one rock each), a drill is needed for hard rocks, and the route SiteC → SiteD is a one-way steep slope. The task is modelled in PDDL, solved with the Fast Downward planner using two search configurations, verified by step-by-step simulation, and then extended with two more rocks and a dust storm that closes Base ↔ SiteB.

2. Modelling

2.1 Domain (Domain.pddl)

The domain uses only **:strips** and **:typing**. It uses no negative preconditions and no numeric fluents.

Types: location, rock, bay.

Predicates: rover-at(?l), connected(?from, ?to) (directed edge), rock-at(?r, ?l), drill-at(?l), has-drill, soft(?r), hard(?r), bay-empty(?b), in-bay(?r, ?b).

Action	Preconditions	Effects
move(?from, ?to)	rover-at(?from), connected(?from, ?to)	add rover-at(?to); delete rover-at(?from)
pick-drill(?l)	rover-at(?l), drill-at(?l)	add has-drill; delete drill-at(?l)
collect-soft(?r, ?l, ?b)	rover-at(?l), rock-at(?r, ?l), soft(?r), bay-empty(?b)	add in-bay(?r, ?b); delete rock-at(?r, ?l), bay-empty(?b)
collect-hard(?r, ?l, ?b)	as collect-soft, with hard(?r) and has-drill	same as collect-soft
deposit(?r, ?l, ?b)	rover-at(?l), in-bay(?r, ?b)	add rock-at(?r, ?l), bay-empty(?b); delete in-bay(?r, ?b)

Design notes:

- Soft and hard collection are separate actions, so the drill requirement is a positive precondition (has-drill). This keeps the model within STRIPS.
- has-drill is never deleted, so once the drill is picked up the rover keeps it for the rest of the mission.
- Each bay holds at most one rock, because collection requires bay-empty and deletes it.
- Deposit is allowed at any location. The goal, which requires all rocks at Base, forces delivery there.
- Each two-way route is encoded as two directed connected facts. The one-way slope is the single fact (connected sitec sited); the reverse edge is absent.

2.2 Original problem (Problem.pddl)

- **Objects:** locations base, sitea to sitef; rocks rock1 to rock5; bays bay1, bay2.
- **Initial state:** rover at Base; drill at SiteE; Rock1 at SiteA, Rock2 at SiteB, Rock3 at SiteC, Rock4 at SiteD, Rock5 at SiteF. Rock3 and Rock5 are hard; the others are soft. Both bays are empty.
- **Connectivity:** as given in the problem brief (two-way routes Base–SiteA, Base–SiteB, SiteA–SiteC, SiteB–SiteD, SiteC–SiteE, SiteD–SiteE, SiteE–SiteF, plus the one-way SiteC → SiteD).
- **Goal:** Rocks 1 to 5 at Base, rover at Base, Bay1 and Bay2 empty.

2.3 Extended problem (Problem_extended.pddl)

The extended problem adds Rock6 (soft, at SiteE) and Rock7 (hard, at SiteF). Both (connected base siteb) and (connected siteb base) are removed to represent the dust storm. The goal requires Rocks 1 to 7 at Base, the rover at Base, and both bays empty.

3. Planner and Search Configurations

The planner is Fast Downward, built from the official GitHub repository and run in Google Colab. Both configurations use unit action costs.

Configuration	Fast Downward option	Type	Guarantee
Satisficing	eager_greedy([ff()]) (greedy best-first search, FF heuristic)	Satisficing	Finds a plan, not necessarily the shortest
Optimal	astar(lmcut()) (A* with the admissible LM-cut heuristic)	Optimal	Shortest plan

4. Results (Task 3)

4.1 Original scenario

Satisficing plan (greedy best-first search, FF), 30 actions:

```
01 (move base sitea)                 11 (move sitec sitee)                21 (move base sitea)
02 (collect-soft rock1 sitea bay1)   12 (pick-drill sitee)                22 (move sitea sitec)
03 (move sitea base)                 13 (move sitee sitec)                23 (move sitec sitee)
04 (deposit rock1 base bay1)         14 (collect-hard rock3 sitec bay1)   24 (move sitee sitef)
05 (move base siteb)                 15 (move sitec sited)                25 (collect-hard rock5 sitef bay1)
06 (collect-soft rock2 siteb bay1)   16 (collect-soft rock4 sited bay2)   26 (move sitef sitee)
07 (move siteb base)                 17 (move sited siteb)                27 (move sitee sitec)
08 (deposit rock2 base bay1)         18 (move siteb base)                 28 (move sitec sitea)
09 (move base sitea)                 19 (deposit rock3 base bay1)         29 (move sitea base)
10 (move sitea sitec)                20 (deposit rock4 base bay2)         30 (deposit rock5 base bay1)
```

Optimal plan (A* + LM-cut), 25 actions:

```
01 (move base siteb)                 10 (move sitea sitec)                19 (move sitea base)
02 (collect-soft rock2 siteb bay1)   11 (move sitec sitee)                20 (deposit rock3 base bay2)
03 (move siteb sited)                12 (pick-drill sitee)                21 (deposit rock5 base bay1)
04 (collect-soft rock4 sited bay2)   13 (move sitee sitef)                22 (move base sitea)
05 (move sited siteb)                14 (collect-hard rock5 sitef bay1)   23 (collect-soft rock1 sitea bay1)
06 (move siteb base)                 15 (move sitef sitee)                24 (move sitea base)
07 (deposit rock2 base bay1)         16 (move sitee sitec)                25 (deposit rock1 base bay1)
08 (deposit rock4 base bay2)         17 (collect-hard rock3 sitec bay2)
09 (move base sitea)                 18 (move sitec sitea)
```

4.2 Extended scenario

Satisficing plan (greedy best-first search, FF), 46 actions:

```
01 (move base sitea)                 17 (move sited siteb)                33 (move sitef sitee)
02 (collect-soft rock1 sitea bay1)   18 (collect-soft rock2 siteb bay1)   34 (move sitee sitec)
03 (move sitea base)                 19 (move siteb sited)                35 (move sitec sitea)
04 (deposit rock1 base bay1)         20 (collect-soft rock4 sited bay2)   36 (move sitea base)
05 (move base sitea)                 21 (move sited sitee)                37 (deposit rock5 base bay1)
06 (move sitea sitec)                22 (move sitee sitec)                38 (deposit rock7 base bay2)
07 (move sitec sitee)                23 (move sitec sitea)                39 (move base sitea)
08 (pick-drill sitee)                24 (move sitea base)                 40 (move sitea sitec)
09 (move sitee sitec)                25 (deposit rock2 base bay1)         41 (move sitec sitee)
10 (collect-hard rock3 sitec bay1)   26 (deposit rock4 base bay2)         42 (collect-soft rock6 sitee bay1)
11 (move sitec sitea)                27 (move base sitea)                 43 (move sitee sitec)
12 (move sitea base)                 28 (move sitea sitec)                44 (move sitec sitea)
13 (deposit rock3 base bay1)         29 (move sitec sitee)                45 (move sitea base)
14 (move base sitea)                 30 (move sitee sitef)                46 (deposit rock6 base bay1)
15 (move sitea sitec)                31 (collect-hard rock5 sitef bay1)
16 (move sitec sited)                32 (collect-hard rock7 sitef bay2)
```

Optimal plan (A* + LM-cut), 40 actions:

```
01 (move base sitea)                 15 (move base sitea)                 29 (collect-hard rock3 sitec bay1)
02 (move sitea sitec)                16 (move sitea sitec)                30 (move sitec sitee)
03 (move sitec sited)                17 (move sitec sitee)                31 (collect-soft rock6 sitee bay2)
04 (collect-soft rock4 sited bay1)   18 (move sitee sitef)                32 (move sitee sitec)
05 (move sited siteb)                19 (collect-hard rock5 sitef bay1)   33 (move sitec sitea)
06 (collect-soft rock2 siteb bay2)   20 (collect-hard rock7 sitef bay2)   34 (move sitea base)
07 (move siteb sited)                21 (move sitef sitee)                35 (deposit rock3 base bay1)
08 (move sited sitee)                22 (move sitee sitec)                36 (deposit rock6 base bay2)
09 (pick-drill sitee)                23 (move sitec sitea)                37 (move base sitea)
10 (move sitee sitec)                24 (move sitea base)                 38 (collect-soft rock1 sitea bay1)
11 (move sitec sitea)                25 (deposit rock5 base bay1)         39 (move sitea base)
12 (move sitea base)                 26 (deposit rock7 base bay2)         40 (deposit rock1 base bay1)
13 (deposit rock2 base bay2)         27 (move base sitea)
14 (deposit rock4 base bay1)         28 (move sitea sitec)
```

4.3 Planner summary

Plan length equals total action cost because every action has unit cost, and the total number of actions in each plan equals its plan length. Expanded-state counts and search times are taken from the four planner output files. Fast Downward reports the search time; the wrapper-reported total planner time (translation plus search) is shown separately.

Scenario	Configuration	Plan length (= total actions)	States expanded	Search time (s)	Total planner time (s)
Original	Greedy (FF)	30	51	0.002297	0.24
Original	A* (LM-cut)	25	5,776	0.499424	0.66
Extended	Greedy (FF)	46	92	0.004023	0.16
Extended	A* (LM-cut)	40	2,230,060	264.084328	264.28

5. Goal Verification (Task 4)

Each plan was verified with a step-by-step STRIPS simulator, **simulate_plans.py** (included in the submission). The simulator does not hard-code the rover rules. It parses Domain.pddl to obtain the action schemas, parses Problem.pddl or Problem_extended.pddl to obtain the initial state and goal, and reads the plan from the planner output file. For every action it checks that every precondition holds in the current state, applies the delete effects and then the add effects, and records the state. At the end it checks that every goal atom is true. The number of actions it simulates is also compared with the plan length reported by Fast Downward. Run **python3 simulate_plans.py** in the folder containing the submitted files to reproduce every table in this section.

The simulator was also tested on deliberately invalid plans, and it rejects all three: travelling SiteD → SiteC against the one-way slope, collecting a hard rock before picking up the drill, and filling a bay that is already full. This shows that it detects violations rather than accepting any plan.

In the tables below, rock numbers in the last column are rock IDs (for example "2, 4" means Rock2 and Rock4 are at Base). All four plans are VALID.

5.1 Original scenario, optimal plan (A*, LM-cut), 25 actions

Initial state: rover at Base, drill at SiteE, both bays empty, no rocks at Base.

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
1	(move base siteb)	SiteB	empty	empty	no	none
2	(collect-soft rock2 siteb bay1)	SiteB	rock2	empty	no	none
3	(move siteb sited)	SiteD	rock2	empty	no	none
4	(collect-soft rock4 sited bay2)	SiteD	rock2	rock4	no	none
5	(move sited siteb)	SiteB	rock2	rock4	no	none
6	(move siteb base)	Base	rock2	rock4	no	none
7	(deposit rock2 base bay1)	Base	empty	rock4	no	2
8	(deposit rock4 base bay2)	Base	empty	empty	no	2, 4
9	(move base sitea)	SiteA	empty	empty	no	2, 4
10	(move sitea sitec)	SiteC	empty	empty	no	2, 4

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
11	(move sitec sitee)	SiteE	empty	empty	no	2, 4
12	(pick-drill sitee)	SiteE	empty	empty	yes	2, 4
13	(move sitee sitef)	SiteF	empty	empty	yes	2, 4
14	(collect-hard rock5 sitef bay1)	SiteF	rock5	empty	yes	2, 4
15	(move sitef sitee)	SiteE	rock5	empty	yes	2, 4
16	(move sitee sitec)	SiteC	rock5	empty	yes	2, 4
17	(collect-hard rock3 sitec bay2)	SiteC	rock5	rock3	yes	2, 4
18	(move sitec sitea)	SiteA	rock5	rock3	yes	2, 4
19	(move sitea base)	Base	rock5	rock3	yes	2, 4
20	(deposit rock3 base bay2)	Base	rock5	empty	yes	2, 3, 4
21	(deposit rock5 base bay1)	Base	empty	empty	yes	2, 3, 4, 5
22	(move base sitea)	SiteA	empty	empty	yes	2, 3, 4, 5
23	(collect-soft rock1 sitea bay1)	SiteA	rock1	empty	yes	2, 3, 4, 5
24	(move sitea base)	Base	rock1	empty	yes	2, 3, 4, 5
25	(deposit rock1 base bay1)	Base	empty	empty	yes	1, 2, 3, 4, 5

Final-state item	Value	Required by the goal?
Rover location	Base	Base: satisfied
Bay1 / Bay2	empty / empty	both empty: satisfied
Rocks at Base	1, 2, 3, 4, 5	Rock1 to Rock5: satisfied
Drill	carried by the rover	not constrained by the goal

Final state (all atoms): (bay-empty bay1) (bay-empty bay2) (connected base sitea) (connected base siteb) (connected sitea base) (connected sitea sitec) (connected siteb base) (connected siteb sited) (connected sitec sitea) (connected sitec sited) (connected sitec sitee) (connected sited siteb) (connected sited sitee) (connected sitee sitec) (connected sitee sited) (connected sitee sitef) (connected sitef sitee) (hard rock3) (hard rock5) (has-drill) (rock-at rock1 base) (rock-at rock2 base) (rock-at rock3 base) (rock-at rock4 base) (rock-at rock5 base) (rover-at base) (soft rock1) (soft rock2) (soft rock4)

Goal check: VALID. Plan length 25; every action's preconditions held and all 8 goal atoms are true in the final state.

5.2 Original scenario, satisficing plan (greedy, FF), 30 actions

Initial state: rover at Base, drill at SiteE, both bays empty, no rocks at Base.

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
1	(move base sitea)	SiteA	empty	empty	no	none
2	(collect-soft rock1 sitea bay1)	SiteA	rock1	empty	no	none
3	(move sitea base)	Base	rock1	empty	no	none
4	(deposit rock1 base bay1)	Base	empty	empty	no	1
5	(move base siteb)	SiteB	empty	empty	no	1
6	(collect-soft rock2 siteb bay1)	SiteB	rock2	empty	no	1
7	(move siteb base)	Base	rock2	empty	no	1
8	(deposit rock2 base bay1)	Base	empty	empty	no	1, 2
9	(move base sitea)	SiteA	empty	empty	no	1, 2
10	(move sitea sitec)	SiteC	empty	empty	no	1, 2
11	(move sitec sitee)	SiteE	empty	empty	no	1, 2
12	(pick-drill sitee)	SiteE	empty	empty	yes	1, 2
13	(move sitee sitec)	SiteC	empty	empty	yes	1, 2
14	(collect-hard rock3 sitec bay1)	SiteC	rock3	empty	yes	1, 2
15	(move sitec sited)	SiteD	rock3	empty	yes	1, 2
16	(collect-soft rock4 sited bay2)	SiteD	rock3	rock4	yes	1, 2
17	(move sited siteb)	SiteB	rock3	rock4	yes	1, 2
18	(move siteb base)	Base	rock3	rock4	yes	1, 2
19	(deposit rock3 base bay1)	Base	empty	rock4	yes	1, 2, 3
20	(deposit rock4 base bay2)	Base	empty	empty	yes	1, 2, 3, 4

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
21	(move base sitea)	SiteA	empty	empty	yes	1, 2, 3, 4
22	(move sitea sitec)	SiteC	empty	empty	yes	1, 2, 3, 4
23	(move sitec sitee)	SiteE	empty	empty	yes	1, 2, 3, 4
24	(move sitee sitef)	SiteF	empty	empty	yes	1, 2, 3, 4
25	(collect-hard rock5 sitef bay1)	SiteF	rock5	empty	yes	1, 2, 3, 4
26	(move sitef sitee)	SiteE	rock5	empty	yes	1, 2, 3, 4
27	(move sitee sitec)	SiteC	rock5	empty	yes	1, 2, 3, 4
28	(move sitec sitea)	SiteA	rock5	empty	yes	1, 2, 3, 4
29	(move sitea base)	Base	rock5	empty	yes	1, 2, 3, 4
30	(deposit rock5 base bay1)	Base	empty	empty	yes	1, 2, 3, 4, 5

Final-state item	Value	Required by the goal?
Rover location	Base	Base: satisfied
Bay1 / Bay2	empty / empty	both empty: satisfied
Rocks at Base	1, 2, 3, 4, 5	Rock1 to Rock5: satisfied
Drill	carried by the rover	not constrained by the goal

Final state (all atoms): (bay-empty bay1) (bay-empty bay2) (connected base sitea) (connected base siteb) (connected sitea base) (connected sitea sitec) (connected siteb base) (connected siteb sited) (connected sitec sitea) (connected sitec sited) (connected sitec sitee) (connected sited siteb) (connected sited sitee) (connected sitee sitec) (connected sitee sited) (connected sitee sitef) (connected sitef sitee) (hard rock3) (hard rock5) (has-drill) (rock-at rock1 base) (rock-at rock2 base) (rock-at rock3 base) (rock-at rock4 base) (rock-at rock5 base) (rover-at base) (soft rock1) (soft rock2) (soft rock4)

Goal check: VALID. Plan length 30; every action's preconditions held and all 8 goal atoms are true in the final state.

5.3 Extended scenario, optimal plan (A*, LM-cut), 40 actions

Initial state: rover at Base, drill at SiteE, both bays empty, no rocks at Base.

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
1	(move base sitea)	SiteA	empty	empty	no	none
2	(move sitea sitec)	SiteC	empty	empty	no	none
3	(move sitec sited)	SiteD	empty	empty	no	none
4	(collect-soft rock4 sited bay1)	SiteD	rock4	empty	no	none
5	(move sited siteb)	SiteB	rock4	empty	no	none
6	(collect-soft rock2 siteb bay2)	SiteB	rock4	rock2	no	none
7	(move siteb sited)	SiteD	rock4	rock2	no	none
8	(move sited sitee)	SiteE	rock4	rock2	no	none
9	(pick-drill sitee)	SiteE	rock4	rock2	yes	none
10	(move sitee sitec)	SiteC	rock4	rock2	yes	none
11	(move sitec sitea)	SiteA	rock4	rock2	yes	none
12	(move sitea base)	Base	rock4	rock2	yes	none
13	(deposit rock2 base bay2)	Base	rock4	empty	yes	2
14	(deposit rock4 base bay1)	Base	empty	empty	yes	2, 4
15	(move base sitea)	SiteA	empty	empty	yes	2, 4
16	(move sitea sitec)	SiteC	empty	empty	yes	2, 4
17	(move sitec sitee)	SiteE	empty	empty	yes	2, 4
18	(move sitee sitef)	SiteF	empty	empty	yes	2, 4
19	(collect-hard rock5 sitef bay1)	SiteF	rock5	empty	yes	2, 4
20	(collect-hard rock7 sitef bay2)	SiteF	rock5	rock7	yes	2, 4
21	(move sitef sitee)	SiteE	rock5	rock7	yes	2, 4
22	(move sitee sitec)	SiteC	rock5	rock7	yes	2, 4
23	(move sitec sitea)	SiteA	rock5	rock7	yes	2, 4
24	(move sitea base)	Base	rock5	rock7	yes	2, 4
25	(deposit rock5 base bay1)	Base	empty	rock7	yes	2, 4, 5

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
26	(deposit rock7 base bay2)	Base	empty	empty	yes	2, 4, 5, 7
27	(move base sitea)	SiteA	empty	empty	yes	2, 4, 5, 7
28	(move sitea sitec)	SiteC	empty	empty	yes	2, 4, 5, 7
29	(collect-hard rock3 sitec bay1)	SiteC	rock3	empty	yes	2, 4, 5, 7
30	(move sitec sitee)	SiteE	rock3	empty	yes	2, 4, 5, 7
31	(collect-soft rock6 sitee bay2)	SiteE	rock3	rock6	yes	2, 4, 5, 7
32	(move sitee sitec)	SiteC	rock3	rock6	yes	2, 4, 5, 7
33	(move sitec sitea)	SiteA	rock3	rock6	yes	2, 4, 5, 7
34	(move sitea base)	Base	rock3	rock6	yes	2, 4, 5, 7
35	(deposit rock3 base bay1)	Base	empty	rock6	yes	2, 3, 4, 5, 7
36	(deposit rock6 base bay2)	Base	empty	empty	yes	2, 3, 4, 5, 6, 7
37	(move base sitea)	SiteA	empty	empty	yes	2, 3, 4, 5, 6, 7
38	(collect-soft rock1 sitea bay1)	SiteA	rock1	empty	yes	2, 3, 4, 5, 6, 7
39	(move sitea base)	Base	rock1	empty	yes	2, 3, 4, 5, 6, 7
40	(deposit rock1 base bay1)	Base	empty	empty	yes	1, 2, 3, 4, 5, 6, 7

Final-state item	Value	Required by the goal?
Rover location	Base	Base: satisfied
Bay1 / Bay2	empty / empty	both empty: satisfied
Rocks at Base	1, 2, 3, 4, 5, 6, 7	Rock1 to Rock7: satisfied
Drill	carried by the rover	not constrained by the goal

Final state (all atoms): (bay-empty bay1) (bay-empty bay2) (connected base sitea) (connected sitea base) (connected sitea sitec) (connected siteb sited) (connected sitec sitea) (connected sitec sited) (connected sitec sitee) (connected sited siteb) (connected sited sitee) (connected sitee sitec) (connected sitee sited) (connected sitee sitef) (connected sitef sitee) (hard rock3) (hard rock5) (hard rock7) (has-drill) (rock-at rock1 base) (rock-at rock2 base) (rock-at rock3 base) (rock-at rock4 base) (rock-at rock5 base) (rock-at rock6 base) (rock-at rock7 base) (rover-at base) (soft rock1) (soft rock2) (soft rock4) (soft rock6)

Goal check: VALID. Plan length 40; every action's preconditions held and all 10 goal atoms are true in the final state.

5.4 Extended scenario, satisficing plan (greedy, FF), 46 actions

Initial state: rover at Base, drill at SiteE, both bays empty, no rocks at Base.

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
1	(move base sitea)	SiteA	empty	empty	no	none
2	(collect-soft rock1 sitea bay1)	SiteA	rock1	empty	no	none
3	(move sitea base)	Base	rock1	empty	no	none
4	(deposit rock1 base bay1)	Base	empty	empty	no	1
5	(move base sitea)	SiteA	empty	empty	no	1
6	(move sitea sitec)	SiteC	empty	empty	no	1
7	(move sitec sitee)	SiteE	empty	empty	no	1
8	(pick-drill sitee)	SiteE	empty	empty	yes	1
9	(move sitee sitec)	SiteC	empty	empty	yes	1
10	(collect-hard rock3 sitec bay1)	SiteC	rock3	empty	yes	1
11	(move sitec sitea)	SiteA	rock3	empty	yes	1
12	(move sitea base)	Base	rock3	empty	yes	1
13	(deposit rock3 base bay1)	Base	empty	empty	yes	1, 3
14	(move base sitea)	SiteA	empty	empty	yes	1, 3
15	(move sitea sitec)	SiteC	empty	empty	yes	1, 3
16	(move sitec sited)	SiteD	empty	empty	yes	1, 3
17	(move sited siteb)	SiteB	empty	empty	yes	1, 3
18	(collect-soft rock2 siteb bay1)	SiteB	rock2	empty	yes	1, 3
19	(move siteb sited)	SiteD	rock2	empty	yes	1, 3

Step	Action	Rover	Bay1	Bay2	Drill	Rocks at Base
20	(collect-soft rock4 sited bay2)	SiteD	rock2	rock4	yes	1, 3
21	(move sited sitee)	SiteE	rock2	rock4	yes	1, 3
22	(move sitee sitec)	SiteC	rock2	rock4	yes	1, 3
23	(move sitec sitea)	SiteA	rock2	rock4	yes	1, 3
24	(move sitea base)	Base	rock2	rock4	yes	1, 3
25	(deposit rock2 base bay1)	Base	empty	rock4	yes	1, 2, 3
26	(deposit rock4 base bay2)	Base	empty	empty	yes	1, 2, 3, 4
27	(move base sitea)	SiteA	empty	empty	yes	1, 2, 3, 4
28	(move sitea sitec)	SiteC	empty	empty	yes	1, 2, 3, 4
29	(move sitec sitee)	SiteE	empty	empty	yes	1, 2, 3, 4
30	(move sitee sitef)	SiteF	empty	empty	yes	1, 2, 3, 4
31	(collect-hard rock5 sitef bay1)	SiteF	rock5	empty	yes	1, 2, 3, 4
32	(collect-hard rock7 sitef bay2)	SiteF	rock5	rock7	yes	1, 2, 3, 4
33	(move sitef sitee)	SiteE	rock5	rock7	yes	1, 2, 3, 4
34	(move sitee sitec)	SiteC	rock5	rock7	yes	1, 2, 3, 4
35	(move sitec sitea)	SiteA	rock5	rock7	yes	1, 2, 3, 4
36	(move sitea base)	Base	rock5	rock7	yes	1, 2, 3, 4
37	(deposit rock5 base bay1)	Base	empty	rock7	yes	1, 2, 3, 4, 5
38	(deposit rock7 base bay2)	Base	empty	empty	yes	1, 2, 3, 4, 5, 7
39	(move base sitea)	SiteA	empty	empty	yes	1, 2, 3, 4, 5, 7
40	(move sitea sitec)	SiteC	empty	empty	yes	1, 2, 3, 4, 5, 7
41	(move sitec sitee)	SiteE	empty	empty	yes	1, 2, 3, 4, 5, 7
42	(collect-soft rock6 sitee bay1)	SiteE	rock6	empty	yes	1, 2, 3, 4, 5, 7
43	(move sitee sitec)	SiteC	rock6	empty	yes	1, 2, 3, 4, 5, 7
44	(move sitec sitea)	SiteA	rock6	empty	yes	1, 2, 3, 4, 5, 7
45	(move sitea base)	Base	rock6	empty	yes	1, 2, 3, 4, 5, 7
46	(deposit rock6 base bay1)	Base	empty	empty	yes	1, 2, 3, 4, 5, 6, 7

Final-state item	Value	Required by the goal?
Rover location	Base	Base: satisfied
Bay1 / Bay2	empty / empty	both empty: satisfied
Rocks at Base	1, 2, 3, 4, 5, 6, 7	Rock1 to Rock7: satisfied
Drill	carried by the rover	not constrained by the goal

Final state (all atoms): (bay-empty bay1) (bay-empty bay2) (connected base sitea) (connected sitea base) (connected sitea sitec) (connected siteb sited) (connected sitec sitea) (connected sitec sited) (connected sitec sitee) (connected sited siteb) (connected sited sitee) (connected sitee sitec) (connected sitee sited) (connected sitee sitef) (connected sitef sitee) (hard rock3) (hard rock5) (hard rock7) (has-drill) (rock-at rock1 base) (rock-at rock2 base) (rock-at rock3 base) (rock-at rock4 base) (rock-at rock5 base) (rock-at rock6 base) (rock-at rock7 base) (rover-at base) (soft rock1) (soft rock2) (soft rock4) (soft rock6)

Goal check: VALID. Plan length 46; every action's preconditions held and all 10 goal atoms are true in the final state.

6. Extended Scenario (Task 5)

Problem_extended.pddl adds Rock6 and Rock7, removes the Base ↔ SiteB route, and sets the goal to Rocks 1 to 7 at Base with the rover at Base. Both configurations solve it: greedy search returns a 46-action plan and A* returns a 40-action optimal plan (Section 4.2).

With Base ↔ SiteB closed, SiteB is connected only to SiteD, so the rover can reach SiteB only through SiteD. In the optimal plan the rover follows:

Base → SiteA → SiteC → SiteD → SiteB → SiteD → SiteE → SiteC → SiteA → Base,

and then makes further trips to SiteF, to SiteC and SiteE, and to SiteA before returning to Base.

7. Plan Analysis (Task 6)

7.1 Satisficing versus optimal plan length

A* with the admissible LM-cut heuristic returns a minimum-length plan. In the original scenario the optimal length is 25 actions and greedy search returns 30, so greedy uses 5 extra actions. In the extended scenario the lengths are 40 and 46, a difference of 6.

Where the extra length comes from. Every plan must contain the same non-movement actions: one collect per rock, one deposit per rock, and one pick-drill. That is 11 actions in the original scenario (5 + 5 + 1) and 15 in the extended scenario (7 + 7 + 1). Plan length is therefore this fixed number plus the number of move actions, so the whole 5-action (original) and 6-action (extended) gap is extra movement (Table 7.1).

Table 7.1. Action counts per plan.

Plan	Plan length	Moves	Collects	Deposits	Pick-drill	Trips from Base
Original, A* (25)	25	14	5	5	1	3
Original, greedy (30)	30	19	5	5	1	4
Extended, A* (40)	40	25	7	7	1	4
Extended, greedy (46)	46	31	7	7	1	5

Extra movement has two causes, which Table 7.2 separates trip by trip:

- **Extra round trips.** Greedy search makes one more trip to Base than A* (4 against 3 in the original scenario, 5 against 4 in the extended one), and several of its trips carry a single rock. Each additional trip costs a fixed overhead of moves to leave and return to Base.
- **Inefficient routes within trips.** Even where the number of rocks per trip is similar, greedy chooses longer routes. For example, in the extended scenario it spends a whole trip (6 moves) fetching the drill and Rock3 alone, while A* picks up the drill at SiteE on the way back from SiteB when the rover has to pass SiteE anyway, at no extra trip cost.

Table 7.2. Trips from Base: route, rocks delivered and number of moves.

Plan	Trip	Rocks delivered	Route	Moves
Original, A* (25)	1	Rocks 2, 4	Base → B → D → B → Base	4
	2	Rocks 3, 5	Base → A → C → E → F → E → C → A → Base	8
	3	Rock 1	Base → A → Base	2
Original, greedy (30)	1	Rock 1	Base → A → Base	2
	2	Rock 2	Base → B → Base	2
	3	Rocks 3, 4	Base → A → C → E → C → D → B → Base	7
	4	Rock 5	Base → A → C → E → F → E → C → A → Base	8
Extended, A* (40)	1	Rocks 2, 4	Base → A → C → D → B → D → E → C → A → Base	9
	2	Rocks 5, 7	Base → A → C → E → F → E → C → A → Base	8
	3	Rocks 3, 6	Base → A → C → E → C → A → Base	6
	4	Rock 1	Base → A → Base	2
Extended, greedy (46)	1	Rock 1	Base → A → Base	2
	2	Rock 3	Base → A → C → E → C → A → Base	6
	3	Rocks 2, 4	Base → A → C → D → B → D → E → C → A → Base	9
	4	Rocks 5, 7	Base → A → C → E → F → E → C → A → Base	8
	5	Rock 6	Base → A → C → E → C → A → Base	6

Greedy best-first search follows the FF heuristic, which estimates the remaining distance to the goal and ignores the cost already spent. It therefore takes whichever step looks closest to finishing, such as fetching the nearby Rock1 first, and does not plan how rocks are paired across trips or how the routes combine. A* must prove that no shorter plan exists, so it finds the combination of pairings and routes

that needs the fewest moves.

Search effort. The optimality guarantee is expensive. In the original scenario A* expanded 5,776 states against 51 for greedy search. In the extended scenario it expanded 2,230,060 states against 92. The search times were 0.499424 s against 0.002297 s (original) and 264.084328 s against 0.004023 s (extended). Greedy search was much faster but did not give the shortest plan, and the cost of optimality grows quickly as the problem gets larger.

7.2 Use of the two storage bays and the drill

Only two rocks can be carried at once, so collection is split across trips to Base. The two bays are interchangeable, so assigning a rock to Bay1 or Bay2 does not change the plan length.

Original optimal plan:

- **Trip 1 (steps 1 to 8):** Rock2 goes into Bay1 at SiteB and Rock4 into Bay2 at SiteD. Both bays are full on the way back, and both are deposited at Base.
- **Trip 2 (steps 9 to 21):** Rock5 goes into Bay1 at SiteF and Rock3 into Bay2 at SiteC. Both bays are full again, and both are deposited at Base.
- **Trip 3 (steps 22 to 25):** only Bay1 is used, for Rock1. This last rock has to be a trip on its own, because Rock1 is the only rock left.

So the optimal plan fills both bays on every trip except the last, which is why it needs only three trips.

The drill is picked up at SiteE at **step 12**, at the start of trip 2 and while both bays are empty. It is needed there because the two hard rocks (Rock5 and Rock3) are collected next. The plan keeps the drill after that point, since has-drill is never deleted. The rover does not need the drill in trip 1, which only involves soft rocks, so the plan correctly leaves the pickup until the route passes SiteE.

In the extended optimal plan the drill is picked up at **step 9** at SiteE. It is picked up on the way back from SiteB while the rover already carries Rock4 and Rock2, so no extra trip is needed to fetch it. The plan then carries Rock4 and Rock2 together, Rock5 and Rock7 together, Rock3 and Rock6 together, and finally Rock1 alone. Both bays are emptied at Base after each loaded trip.

7.3 Reaching SiteB and SiteD after the dust storm

After the dust storm the direct Base ↔ SiteB connections are removed. SiteB is then reachable only from SiteD, and SiteD can be reached from SiteC (one-way slope) or from SiteE (two-way route).

In the extended optimal plan the rover travels Base → SiteA → SiteC → SiteD → SiteB. The one-way slope SiteC → SiteD is used at step 3 and gives the shortest way to SiteD. On the return it goes SiteB → SiteD → SiteE → SiteC → SiteA → Base, because the reverse edge SiteD → SiteC does not exist. The rover reaches SiteC through SiteE instead, which also takes it past the drill.

The extended greedy plan also uses the slope (action 16) and also returns from SiteD via SiteE and SiteC. So in the extended scenario the one-way route is used in both plans, and always in its permitted direction.

In the original scenario the optimal plan never uses the slope, since SiteB and SiteD are reached directly from Base through SiteB. Only the greedy plan uses it (action 15).

8. Conclusion

The rover problem was modelled in STRIPS PDDL with typing only and solved with greedy best-first search (FF) and A* with LM-cut. All four plans were verified by simulation to end with all rocks at Base, both bays empty, and the rover at Base. In the original scenario greedy search found a 30-action plan against the 25-action optimum; in the extended scenario the lengths were 46 and 40. The gap comes from greedy search making extra, partly empty trips, while A* pairs rocks to fill both bays. Greedy search expanded far fewer states and ran much faster, but gave no quality guarantee. The dust storm forces the rover to reach SiteB through SiteD, and the one-way slope SiteC → SiteD makes that approach shorter without ever being needed in reverse.

Appendix A. Submitted files

- Domain.pddl, Problem.pddl, Problem_extended.pddl
- Planner outputs: planner_output_base_satisficing.txt, planner_output_base_optimal.txt, planner_output_extended_satisficing.txt, planner_output_extended_optimal.txt
- simulate_plans.py: the STRIPS simulator used for Task 4
- Mars_Rover_PDDL_Planning_Report.pdf: this report

Appendix B. Planner output excerpts

Verbatim lines copied from the four planner output files (command line, result and statistics). The complete logs are in the submitted planner_output_*.txt files.

Original scenario, greedy (FF) (planner_output_base_satisficing.txt)

```
INFO  search command line string: /content/downward/builds/release/bin/downward --search 'eager_greedy([ff()])' --inte
    rnal-plan-file plan_base_satisficing.txt < output.sas
[t=0.006706s, 11728 KB] Solution found!
[t=0.006764s, 11728 KB] Plan length: 30 step(s).
[t=0.006764s, 11728 KB] Plan cost: 30
[t=0.006764s, 11728 KB] Expanded 51 state(s).
[t=0.006764s, 11728 KB] Reopened 0 state(s).
[t=0.006764s, 11728 KB] Evaluated 121 state(s).
[t=0.006764s, 11728 KB] Generated 183 state(s).
[t=0.006764s, 11728 KB] Dead ends: 0 state(s).
[t=0.006764s, 11728 KB] Search time: 0.002297s
[t=0.006764s, 11728 KB] Total time: 0.006764s
INFO  Planner time: 0.24s
```

Original scenario, A* (LM-cut) (planner_output_base_optimal.txt)

```
INFO  search command line string: /content/downward/builds/release/bin/downward --search 'astar(lmcut())' --internal-p
    lan-file plan_base_optimal.txt < output.sas
[t=0.502627s, 12144 KB] Solution found!
[t=0.502708s, 12144 KB] Plan length: 25 step(s).
[t=0.502708s, 12144 KB] Plan cost: 25
[t=0.502708s, 12144 KB] Expanded 5776 state(s).
[t=0.502708s, 12144 KB] Reopened 1 state(s).
[t=0.502708s, 12144 KB] Evaluated 11707 state(s).
[t=0.502708s, 12144 KB] Generated 23614 state(s).
[t=0.502708s, 12144 KB] Dead ends: 0 state(s).
[t=0.502708s, 12144 KB] Search time: 0.499424s
[t=0.502708s, 12144 KB] Total time: 0.502708s
INFO  Planner time: 0.66s
```

Extended scenario, greedy (FF) (planner_output_extended_satisficing.txt)

```
INFO  search command line string: /content/downward/builds/release/bin/downward --search 'eager_greedy([ff()])' --inte
    rnal-plan-file plan_extended_satisficing.txt < output.sas
[t=0.008026s, 11728 KB] Solution found!
[t=0.008086s, 11728 KB] Plan length: 46 step(s).
[t=0.008086s, 11728 KB] Plan cost: 46
[t=0.008086s, 11728 KB] Expanded 92 state(s).
[t=0.008086s, 11728 KB] Reopened 0 state(s).
[t=0.008086s, 11728 KB] Evaluated 254 state(s).
[t=0.008086s, 11728 KB] Generated 369 state(s).
[t=0.008086s, 11728 KB] Dead ends: 0 state(s).
[t=0.008086s, 11728 KB] Search time: 0.004023s
[t=0.008086s, 11728 KB] Total time: 0.008086s
INFO  Planner time: 0.16s
```

Extended scenario, A* (LM-cut) (planner_output_extended_optimal.txt)

```
INFO  search command line string: /content/downward/builds/release/bin/downward --search 'astar(lmcut())' --internal-p
    lan-file plan_extended_optimal.txt < output.sas
[t=264.087916s, 212000 KB] Solution found!
[t=264.088011s, 212000 KB] Plan length: 40 step(s).
[t=264.088011s, 212000 KB] Plan cost: 40
[t=264.088011s, 212000 KB] Expanded 2230060 state(s).
[t=264.088011s, 212000 KB] Reopened 3932 state(s).
[t=264.088011s, 212000 KB] Evaluated 3734017 state(s).
[t=264.088011s, 212000 KB] Generated 8993020 state(s).
[t=264.088011s, 212000 KB] Dead ends: 0 state(s).
[t=264.088011s, 212000 KB] Search time: 264.084328s
[t=264.088011s, 212000 KB] Total time: 264.088011s
INFO  Planner time: 264.28s
```